



PROFESSIONAL
EDUCATION

Digital Plus⁺
Programs

Cloud & DevOps: Continuous Transformation



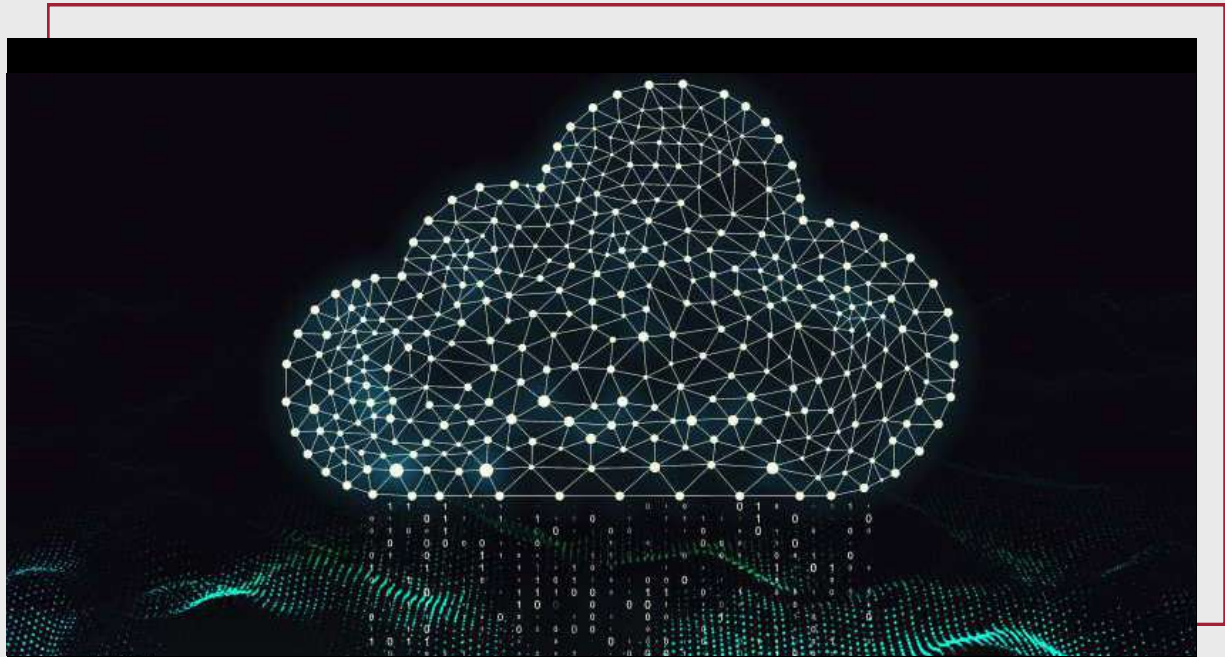
The Cloud is the most important platform in the market today: it is the place where the new generation of digital products will be born.

Dr. Abel Sanchez

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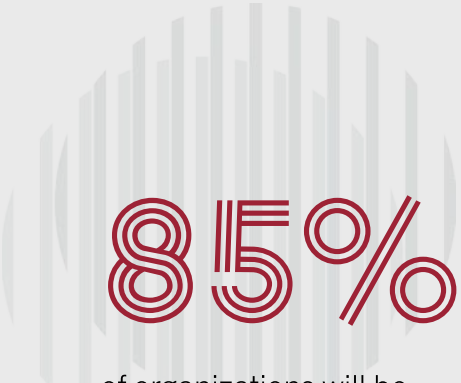
Digital Transformation is here



The digital revolution has disrupted entire industries, transforming the way organizations, companies, and individuals do business. When the infrastructure that supports technology undergoes a fundamental structural change, the future of a business or project depends on understanding these transformations and being able to make strategic decisions that take new technological realities into account.

Two of the most important drivers of digital transformation are Cloud Computing and DevOps, which are already proving to have a profound impact on job markets, organizational culture, technological practices, and corporate strategy. The combined use of Cloud Computing and DevOps methodologies enables organizations to achieve greater agility, scalability, security, and cost-efficiency in their software development and operations.

Cloud & DevOps

A circular graphic with a background of vertical bars of varying heights, creating a bar chart effect.

85%

of organizations will be adopting a cloud computing strategy by 2025

Source: Gartner, 2021

A circular graphic with a background of wavy lines, creating a water-like effect.

50%

of organizations are either hiring new staff or re-training existing staff to better optimize their cloud spend.

Source: Forrester, 2022

A circular graphic with a background of concentric circles, creating a ripple effect.

99%

of professionals report that DevOps has positively impacted their organization.

Source: Atlassian, 2020

About this Course



To better prepare professionals for this new **Cloud-based world**, MIT Professional Education has designed the online course **Cloud & DevOps: Continuous Transformation**. Tackle the intricacies of technologies from a practical standpoint to understand the numerous possibilities and applications that are offered when a **sympiosis of Cloud Computing and DevOps** is properly employed.

During the course, you will:

1

Master the tools for web and Cloud Computing

2

Gain hands-on practice with Git and GitHub, Docker, Node, and NPM

3

Create accounts on DockerHub and Amazon Web Services

4

Analyze a variety of real-world case studies



8 weeks



8-10 hours a week



Online

Key Takeaways

This course employs **MIT's signature hands-on approach** to Cloud Computing and DevOps with a **foundation steeped in theory**. You will gain technical skills with a two-pronged approach including **theoretical and application-oriented perspectives**. Study the mechanisms and tools to take advantage of Cloud Computing and DevOps as secure and affordable software design and efficient technological operations.

After completing this course, you will:

- 01** Become familiar with the **past, present, and future of Cloud Computing** and its impacts on **new developments in technology and business**.
- 02** Gain immediately applicable experience with the **key tools for effective software design, Cloud Computing, and DevOps methodologies**.
- 03** Understand the technicalities of **containers, orchestration, dockers, and KPIs** and how these tools are best applied in today's work environment.
- 04** Optimize both **security and performance in software design** with a **robust and scalable cloud infrastructure**.
- 05** Distinguish the **advantages and disadvantages of different applications**, such as **Serverless and Cloud Native**, and how to leverage them.
- 06** Identify and implement the appropriate approaches to **develop and streamline Cloud Computing and DevOps projects**.



Participant Profile

This course is applicable to all industries and sectors, as well as many distinct professional roles and companies. This course is intended for a variety of professionals, such as:

CTOs, CINOs and leaders of technological departments or projects who need to understand the new possibilities that Cloud Computing offers for digital transformation and advancements in technical areas.

Technical professionals who want to obtain a broader strategic vision about how Cloud Computing and DevOps can transform their business operations.

CEOs, entrepreneurs and functional managers in technical areas or sectors who want to create new business models and upscale their current operations.

Technical prerequisites:

The computer or laptop used for the course must have been purchased or manufactured within the last three years to be compatible with the required software.

It does not have to be owned or managed by a company, as participants will be asked to download software and change computer settings throughout the course.

It should have one of the following operating systems:

- Windows 10 Pro (highly recommended)
- Mac OSX
- Linux



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Instructors



Dr. Abel Sanchez

Executive Director of MIT's Geospatial Data Center, Research Scientist; Center for Complex Engineering Systems, Sociotechnical Systems Research Center, under the Schwarzman School of Computer Science

Dr. Sanchez holds a Ph.D. from the Massachusetts Institute of Technology (MIT). He is the Executive Director of MIT's Geospatial Data Center (GDC). His areas of specialty include the Internet of Things (IOT), Big Data, Cybersecurity, and Digital Innovation. He teaches graduate courses in Data Science, Cybersecurity, and Innovation. For the past eight years his research has focused on architecting large-scale computation.

In IOT, Dr. Sanchez led the global network architecture for the Internet of Things at MIT. The design addresses large-scale computation. Compared to the largest numbers in the world, 2 billion computers, 7 billion phones, and 7 billion people, IOT is orders of magnitude bigger. In similar work,

Dr. Sanchez directed simulation of the U.S. critical infrastructure with the National Infrastructure Simulation and Analysis Center (NISAC).ia.

In data science, early work included supply chain information engineering, analytics, simulation, and visualization with Wal-Mart, Kraft, and SAP. Dr. Sanchez extended his work to global anti-counterfeiting efforts with Johnson & Johnson, SAP, and Altria. Helping Altria scale track-and-trace using the Electronic Product Code Information Services standard from RFID. The work produced a global infrastructure used by industry and government to this day.

In enterprise computing, Dr. Sanchez led the design of a global data infrastructure simulator, modeling follow-the-sun engineering, to evaluate the impact of competing architectures on the performance, availability and reliability of the system for Ford Motor Company. The simulator modeled user actions, applications, background processes, network load, servers, storage, and global data centers. The work identified data center reductions opportunities estimated at a billion dollars in savings.

In cyber security, Dr. Sanchez directed impact analysis of large-scale cyber-attacks designing Cyber Ranges for the Department of Defense (DOD). Conducting repeatable experiments in impact analysis and the ability to model the cyber environment in a highly portable fashion. Looking at the insider threat Dr. Sanchez led the DOD security study on Enterprise Resource Planning systems across the United States Armed Forces.

In password security, Dr. Sanchez led the design of a password firewall (negative authentication) for the Intelligence Advanced Research Projects Activity (IARPA) agency.

The Negative Filtering or Negative Authentication (NA) approach utilizes a form of complement profiles which resembles the censoring and maturation process of T- cells in the immune system.

In machine learning, addressing financial fraud, Dr. Sanchez designed a situational awareness framework that exploits different perspectives of the same financial data and assigns risk scores to entities (e.g. payment documents) to improve false positive ratios and assist the identification of fraudulent activity in huge and unlabeled financial data in collaboration with Accenture.

In physical security, Dr. Sanchez is developing algorithms to assess risk in the integration of information technology (IT) and operations technology (OT).

Dr. Sanchez is the founder and Chief Software Architect of the Open Source RFID platform project. Dr. Sanchez' software systems are used by Samsung, NEC, NTT, Hitachi, Motorola, SAP, IBM, and Microsoft. Other software initiatives are in use by Sandia National Laboratories, MIT, and by several organizations in East Asia and Europe.

Dr. Sanchez advises companies in the Americas, Europe, Middle East, and Asia.

"We use the term 'applied' because we recognize this as a business problem, but we also wanted to ground the technology and give people fundamentals."

Affiliations

- MIT Geospatial Data Center (GDC)
- MIT Sociotechnical Systems Research Center (SSRC)
- MIT Institute for Data, Systems, and Society (IDSS)
- MIT Center for Complex Engineering Systems (CCES)
- MIT Center for Computational Science and Engineering (CCSE)
- MIT Consortium for Improving Critical Infrastructure Cybersecurity (IC3)
- MIT Schwarzman School of Computer Science
- MIT AutoID Laboratory

Areas of Interest and Expertise

- Machine Learning
- Cyber/Physical Security
- Enterprise Computing
- Data Science
- IOT

Courses Taught with MIT Professional Education

- Applied Generative AI for Digital Transformation
- Blockchain: Disruptive Technology
- Cloud & DevOps: Continuous Transformation
- Data Leadership: Transforming the Corporation's Operations, Management, and Mindset to Leverage Data, AI, and Cloud Computing
- Digital Transformation: From AI and IOT to Cloud, Blockchain, and Cybersecurity



Prof. John R. Williams

Professor of Information Engineering, Civil and Environmental Engineering and Director of MIT Geospatial Data Center, and a faculty member in the Center for Computational Science and Engineering part of the Schwarzman School of Computer Science

John Williams holds a BA in Physics from Oxford University, a MS in Physics from UCLA, and a Ph.D. in Numerical Methods from University of Wales, Swansea. His research focuses on the application of Exascale computation to problems in cyber-physical systems, security and energy. His work on fault-tolerant computing using container migration won the IEEE, High Performance Extreme Computing Conference award for Best Innovation in 2019. He is director of MIT's Geospatial Data Center and from 2006-2012, was Director of the MIT Auto-ID Laboratory, where the Internet of Things was invented.

He is author or co-author of over 250 journal and conference papers, as well as the books on Rock Mechanics and RFID Technology. He contributed to the 2013 report for the UK Office for Science Foresight Project- The Future of Manufacturing. Alongside Bill Gates and Larry Ellison, he was named as one of the 50 most powerful people in Computer Networks. He consults to companies including Accenture, Schlumberger, Shell, Total, Exxon, SAP Research, Microsoft Research, Kajima Corp, US Lincoln Laboratory, Sandia National Laboratories, US Intelligence Advanced Research Projects Activity, Motorola, Phillip-Morris Inc., Ford Motor Company, Exxon-Mobil, Shell, Total, and ARAMCO.

His international collaborations include Oxford and Cambridge Universities, HKUST, KACST, Alfaisal University, PolyU Hong Kong, Imperial College of Science and Technology UK, Malaysia University of Science and Technology (MUST), and Masdar Institute of Science and Technology Abu Dhabi. He organized the first Cyber-Physical Security Conference in the UK (2011) and along with Dr. Sanchez, he runs the MIT Applied Cyber Security Professional Education summer course. At MIT he teaches courses Architecting Software Systems (MIT 1.125) and Engineering Computation and Data Science (MIT 1.00/1.001).

In data engineering and data science, early work included simulation of Ford's global network, analysis of SAP smart grid billing system. For Altria, he analyzed the performance of item level tagging and also their implementation of an anti-counterfeiting system using the Electronic Product Code (EPC).

In password security, Dr. Williams was PI that developed the algorithms for a negative password authentication system for the Intelligence Advanced Research Projects Activity (IARPA) agency.

Dr. Williams advises companies in the Americas, Europe, Middle East, and Asia.

“Data engineering builds the corporate nervous system and AI is going to make its reactions smart.”

Affiliations

- MIT Department of Civil and Environmental Engineering
- MIT Center for Computational Science and Engineering (CCSE)
- MIT Schwarzman School of Computer Science
- MIT Geospatial Data Center (GDC)
- MIT Auto-ID Laboratory
- MIT Center for Complex Engineering Systems (CCES)
- MIT Consortium for Improving Critical Infrastructure Cybersecurity (IC3)
- MIT System Design and Management Program

Areas of Interest and Expertise

- Information Technology,
- Cyber/Physical Security
- Web-Based Education Technology
- Large Scale Network Simulation
- GeoNumerics of Granular and Powder Systems
- Modern Software Architecting and Cyber Security
- Web Services and Distributed Computing
- Discrete Element Simulation and Analysis of Discontinua

Courses Taught with MIT Professional Education

- Applied Generative AI for Digital Transformation
- Blockchain: Disruptive Technology
- Cloud & DevOps: Continuous Transformation
- Data Leadership: Transforming the Corporation's Operations, Management, and Mindset to Leverage Data, AI, and Cloud Computing
- Digital Transformation: From AI and IOT to Cloud, Blockchain, and Cybersecurity

Course Outline

Module 1

Into the Cloud: How We Got Here

In this module, you will learn to describe the basic problems of transversal computing in terms of Remote Procedure Calls (RPC) and the management of data flows. You will be able to explain how the HTTP, HTML, and URL specifications, created by Sir Tim Berners-Lee, helped solve these problems. Finally, you will be able to describe the different generations of the web and the importance of smartphones as a means of collecting data.

1. Into the Cloud: The Story of How
2. Into the Cloud: The Story of How We Got to the Cloud
3. The Rise of the Internet and the World Wide Web
4. Web 2.0 and Going Mobile
5. Mobile and IoT: Everyone Generating Data
6. Cloud: The New Normal

Module 2

Web Computing, Open-Source Software, and the Cloud

In this module, you will learn how to develop a web server that you can control from your browser to send data at your request. This data can be found in files or generated from machines or devices (eg. sensors). We will focus on the software known as Node.js, which supports asynchronous computing by the web server, which makes it possible to execute various tasks in parallel.

1. Computing Across Machines at Scale
2. The Browser – A Modern Data Portal
3. The Open-Source Movement
4. The Barriers to Entry Lower

Module 3

Containers and Keys

In this module, you will learn about containers and how to create and run images. In addition, we will raise the issue of container orchestration, which allows us to monitor, extend, manage, and synchronize large numbers of containers. Since cloud computing requires security, we will use keys to communicate with cloud services, for example, GitHub and Docker Hub. Likewise, we will study how the so-called Public Key Infrastructure (PKI) work, their public keys, and their private keys.

1. What are Containers?
2. What is Docker?
3. How is Docker Used?
4. PKI and Keys

Module 4

DevOps

In this module you will learn to identify the main elements of DevOps and the flow of information. You will know how to apply DevOps and how this same practice can contribute to increasing efficiency and improving the allocation of resources in those processes. You'll also learn about the typical metrics of high-performing companies, listing the four most important ones: Wait Time, Deployment Frequency, Service Restoration Time, and Failure Rate.

1. What is DevOps?
2. What Does It Mean to Be Agile?
3. How DevOps Has Transformed Software
4. Implementing DevOps

Module 5

Case Studies

In this module, you will look at four different case studies of companies that have moved to the cloud. The first case, Microsoft testing in the cloud, is a good example of moving from a traditional work model to a cloud-based work model. Also, another good example of disruption is the case of the Netflix monkeys, who chose an innovative perspective to join the cloud. On the other hand, you will also see a failed case of cloud implementation, that is, the example of the failed transformation of General Electric, to try to understand why this failed. The module will conclude with a case study of Amazon Web Services (AWS).

1. Organizational Transformation and Stages of Adoption
2. Microsoft Testing in the Cloud
3. General Electric Case Study
4. Netflix Case Study
5. AWS Case Study – Move to the Cloud Recommendations
6. Architecting Software in the Cloud
7. The OODA Loop

Module 6

Serverless

Serverless computing is a cloud execution model in which a provider activates the server and dynamically manages the allocation of resources to machines. According to AWS, serverless is the cloud-native architecture that allows us to delegate our operational responsibilities to the cloud provider. In turn, this makes it possible to increase our agility, speed of implementation, and ability to innovate. Examples of serverless applications include warehouses, chatbots, natural language processing, image recognition, databases, API gateways, messaging services, and expressions (Lambda).

1. What is Serverless Computing?
2. 'Going Serverless' - Pros and Cons
3. Function as a Service (FaaS): AWS Lambda
4. Serverless Security Setup Example
5. Writing a Serverless Lambda Function
6. Serverless Architecture for a BadBank

Module 7

The Agile Corporation

The goal of this module is not to design the perfect product, but to create a product to know before the competition what that level of perfection could be. In this module, we will discuss what it means to be Agile and how cloud promotes and makes this goal easier to achieve.

1. The OODA Loop
2. What Does It Mean to Be an Agile Company?
3. The Traditional Corporation's Increasing Lack of Agility
4. How Can You Become Agile?
5. Unblocking Innovation for Digital Transformation
6. Stigmergy – Leave a Trail

Module 8

Cloud Native

For an application to be considered cloud native, it must have the following characteristics: Be fault tolerant. Can be expanded horizontally. Be written in a way that takes advantage of everything cloud providers have to offer. In this last module of the program, what it really means to use cloud native computing and how any company that wants to do so can migrate to the cloud will be analyzed.

1. What is Cloud Native Computing?
2. What Is Kubernetes and How Is It Used?
3. Why Should You Care About Cloud Native Computing?
4. The Trail Map to Cloud Native

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Certificate of Completion

All participants who successfully complete the course will receive an MIT Professional Education Certificate of Completion. In addition, they will also earn 7.8 Continuing Education Units (MIT CEUs). *

In order to obtain CEUs, participants must complete a required CEU confirmation form. CEUs are calculated based on the number of learning hours in each course.



The Continuing Education Unit (CEU) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.



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Welcome to the gateway

to MIT knowledge
and expertise for
professionals
around the world.

155

Participants from
155 countries

≈ 1600

More than 1,600
online activities

92%

92% of Professional
Alumni rated
their experience
as outstanding

About MIT Professional Education

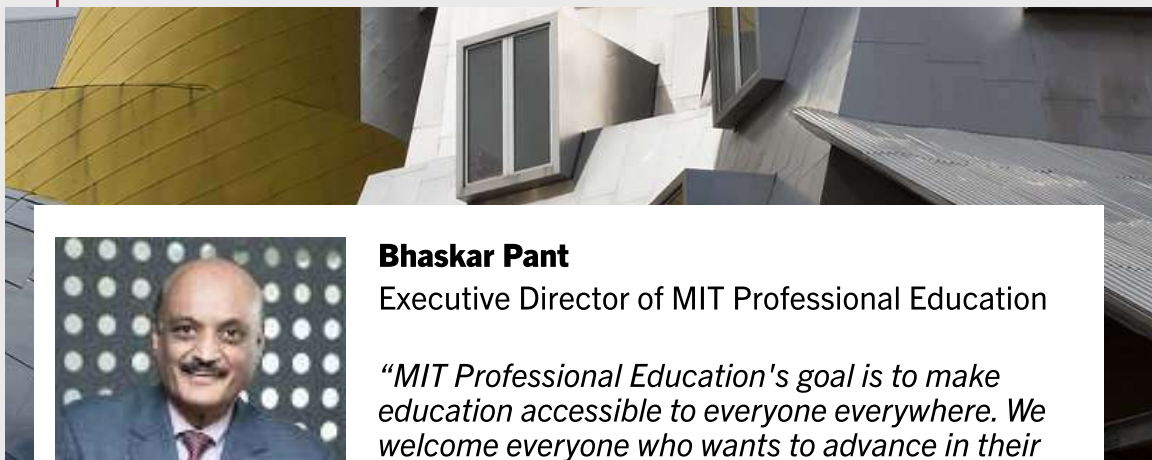
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind**.



Bhaskar Pant

Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:

Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs:

Custom-made to meet the specific needs of each organization and its professionals.

3

Private Cohorts:

Available in Open Enrollment courses with a minimum of participants.



For more information on our Programs for Organizations, please contact us at: **professionalprograms@mit.edu**

The Beyond Online Methodology

MIT Professional Education is **revolutionizing the digital learning experience**. In fact, we are no longer talking about online-only, but rather an **interactive and collaborative learning experience** that's digitally powered: a **Beyond Online experience**.

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology. You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Attend select live webinars with MIT Instructors.
- Collaborate and engage with world-renowned MIT Instructors.
- Learn from expert learning facilitators, guest speakers, and subject matter experts from industries.
- Weekly live webinars from learning facilitators.
- Fulfill your reskilling and upskilling needs in this fast-changing environment.
- Master the technical and human skills necessary for effective leadership.
- Meet and network with other program participants from all around the world.



Clara Piloto

Director of Global Programs at MIT Professional Education

"MIT Professional Education's programs tackle all the technologies that are at the heart of the digital transformation, enabling dedicated professionals to meet new challenges in the Fourth Industrial Revolution and lead change within their organizations"

Benefits of Joining the Alumni Community



MIT Professional Education offers **a number of benefits for participants** who successfully complete our courses:

- 1 **15% tuition reduction** for MIT Professional Education online courses and short-duration, in-person programs.
- 2 **Continued access to the virtual campus for six months after completion** of the course.
- 3 **Exclusive announcements** of new courses, programs, and events through our **virtual Alumni Campus: Professional Network**.
- 4 Invitation to the **MIT Professional Education Alumni LinkedIn group**.
- 5 **Networking opportunities** with other MIT Professional Education Alumni.



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Are you ready to master the new paradigm of the 21st century?

Frequently Asked Questions

global  alumni
-excellence or nothing-

For more information, please contact us:

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